

DEPARTMENT OF COMPUTER SCIENCE & IT

Project Name	Scheduling Algorithms (FCFS and Round Robin)
Course	CS222L - Introduction to Operating Systems Lab (3+1)
Teacher	Sir Waqas Akber
Program	BS Computer Science / Information Technology
Group Members	2020F-BCS-152 – Muhammad Faisal 2020F-BCS-162 – Saad Ali 2020F-BCS-180 – Hamza Khan 2020F-BCS-191 – Muhammad Khubaib
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Abstract

The project aims to implement First Come First Serve (FCFS) and Round Robin (RR) CPU scheduling algorithms using a C# Windows Form GUI and SQL Server for data management. The application provides an intuitive interface for user authentication, process management, and metric calculation. Additional premium features include Gantt Chart visualization and detailed breakdowns of calculations. This project enhances learning and provides insights into CPU scheduling, catering to a wide range of users through both free and paid features.

Software / Hardware / Language / Tool Used

- **Programming Language:** C#
- **Framework:** .NET Framework for Windows Forms
- **Database:** SQL Server
- **GUI Design:** Windows Forms

Benefits

- **Educational:** Provides a comprehensive understanding of FCFS and RR scheduling algorithms.
- **Practical Implementation:** Demonstrates real-world application of scheduling algorithms with user-friendly GUI.
- **User Interaction:** Allows users to input and manage process data, calculate and display scheduling metrics.
- **Enhanced Features:** Premium users benefit from Gantt Chart visualizations and detailed calculation steps.
- **Feedback Mechanism:** Collects user feedback for continuous improvement.